Before it is too late: primary school students and their mental health challenges in Southern Thailand

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Resumo (Abstract)

Purpose To determine the mental health problems and well-being of primary school students in southern Thailand. Methods A survey was conducted from 2019 to 2020 among first- to third-grade students (aged 6–9 years) in southern Thailand. Parents completed self-report questionnaires, including questions on socioeconomic characteristics, family functioning, use of information and technology devices, screen time, and mental health problems before and after the child entered kindergarten. All the students completed the happiness, self-esteem, and life-satisfaction questionnaire. Parents and teachers voluntarily reported perceived mental health concerns. School achievement was assessed by students' grade point averages (GPAs). Results Of the 2,499 students, 34.5% reported poor family functioning, 18.2% had an increased risk of game addiction, and 7.2% already had game addiction. The average screen time (mean ± SD) was 3.5 ± 3.1 h on weekdays and 6.2 ± 4.1 h on weekends, respectively. Students had significantly longer screen time if they owned a mobile phone, had unrestricted internet access, had a desktop computer at home, or had access to more than one device. Approximately one-third of the students had at least one mental health problem, especially academic-related problems. One-fourth of them had a GPA below 2.50. Almost all the students had moderate to high self-esteem. 10% felt sad about themselves, and approximately one-third had poor to very poor life satisfaction. Conclusion Significant mental health challenges coexist with positive indicators of well-being, such as self-esteem and happiness. Appropriate intervention is needed to reduce the problems and promote their resources. Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00127-025-03033-9 .

Introduction

The Global Burden Study in 2024 reported a 6.8% prevalence of mental health disorders among children aged 5–9 years, and gradually increased to 14.2% among children aged 10–14 years [1 – 4]. Depression is among the leading causes of illness and disability among adolescents. Suicide is the third leading cause of death in 15–19-year-olds [4]. These mental health disorders take time to develop and stem from many risk factors long before they reach their teens, hence they must be identified early.

The mental health of young children is increasingly critical, with an increasing number of issues affecting their development, contrary to the assumption that young children are less likely to experience significant mental health problems [5 , 6]. Many Asian countries—including Malaysia, Indonesia, Vietnam, China, and South Korea—have consistently reported rising trends in depression, anxiety, ADHD, and behavioral disorders among children and adolescents [7 – 10]. Rapid urbanization, high academic pressure, digital media exposure, and shifting family structures have been repeatedly identified as driving factors behind these increases [11 , 12]. Although prevalence varies across countries due to differences in assessment and cultural interpretations, a consistent pattern emerges, our children face increasing risks of emotional distress and internalizing problems [13]. Studies suggest that these challenges, if unaddressed, have long-term implications for both

mental health and academic outcomes [14]. Although several regional initiatives—such as school-based programs in Japan and South Korea and community-based mental health services in Southeast Asia—have aimed to improve access and early detection, primary school-aged children remain an underrepresented group in research and intervention efforts across the region [15].

Southern Thailand has unique social, cultural, and environmental factors that may influence children's mental well-being. Previous studies reported increasing rates of emotional and behavioral problems among school-aged children, particularly related to stress, family instability, socio-economic challenges, and prolonged regional conflict in certain border areas [16 – 18]. Limited access to child mental health services, shortages of trained mental health professionals, and persistent stigma further contribute to significant unmet mental health needs in the region [19].

Traditional approaches to understanding child and adolescent mental health have often focused on adverse outcomes; however, this perspective is insufficient. Children should also experience positive states, including happiness, confidence, and high self-esteem. Understanding both mental challenges and resources of young children is helpful to develop a mental health promotion intervention for young children, especially at the school or community level. This study is the first part of our school-based mental health promotion program. It aims to determine the prevalence of mental health problems, self-esteem, happiness, life satisfaction, and their associated factors among first- to third-grade students in southern Thailand.

Materials and methods

Study design and setting

A mixed-methods approach was implemented in southern Thailand between 2019 and 2020. This study used two primary data collection methods: a school-based survey and direct observation, which involved obtaining reports from parents and teachers about the children's perceived emotional and behavioral problems.

Sampling method

To ensure generalizability, the schools were stratified by all three Thai school sectors in the study area, namely public, municipal, and private schools in both rural and urban areas. The school selection criteria included medium (121–500 students) and large (501–1500 students) primary schools under the Office of the Basic Education Commission (OBEC). We randomly selected schools within a 30-kilometer radius of the research center to facilitate a subsequent school-based mental health promotion program that required regular visits.

The sample size was estimated based on a 0.7% prevalence rate of mental health problems [9] and a precision of 0.025; the minimal required sample size was 1,291. After accounting for a design effect of 1.5 and a 10% non-response rate, the required sample size increased to 2,421. Eventually, we invited approximately 2,500 students and their parents to participate in the study. All students in the first to third grades in the selected schools were eligible to participate in the study. No exclusion criteria were applied to examine the real situation of mental health among students who could attend school.

Data collection and tools

First, the principals of the selected schools were invited to a meeting. The researchers explained the rationale, details, and implications of the study. The researchers then invited all the schools to participate in the study. The principals of the schools who were willing to participate in the study signed informed consent forms and permitted researchers to conduct the study.

All first- to third-grade students from the selected schools were eligible to participate in the study. Researchers sent an invitation letter to their parents to attend a meeting at their child's school. After the researchers explained the study details, they obtained written informed consent from the parents, to allow their children to participate. The parents or other primary caregivers were subsequently asked to complete a 30-minute self-administered questionnaire on their sociodemographic data, family function, screening for mental health problems, students' use of information technology, and a screening battery for game addiction.

Next, we invited parents and teachers to voluntarily report their concerns about their children's emotional, behavioural, and academic problems in the open-ended questionnaire. While parents reported on their own child, teachers could report any specific child in their classrooms, which had an average of 35–40 students. Researchers invited students whose parents had given consent for them to voluntarily complete a picture-based questionnaire on self-esteem, happiness, and life satisfaction. Grade point average (GPA) data were obtained from the teachers to assess academic achievement.

Variables and measurement

Family function was assessed via the 36-item Chulalongkorn Family Inventory [9], which measures seven dimensions of family function (problem solving, communication, roles, affective responsiveness, affective involvement, behavior control, and general functioning) on the basis of the McMaster Model of Family Functioning (MMFF) [20]. Its internal consistency was 0.88. The Diagnostic and Statistical Manual of Mental Disorders, 5th Edition (DSM-5) [21] Screening questions were used to assess developmental delay and difficulties in sleep, eating, and excretion before and after students entered kindergarten. Current information technology use by their students and screening for game addiction were measured via the Game Addiction Screening Test (GAST) [22], and game addiction protection was measured via the Game Addiction Prediction Scale (GAME-P) [23].

Once the teachers confirmed a convenient date and time, the researchers revisited each school to assess students' self-esteem via the Lifespan Self-Esteem Scale (LSE) [24] and the Rosenberg Self-Esteem Scale (Thai version) [25]. The Cronbach's alpha of the Lifespan Self-Esteem Scale (LSE) is 0.80 [26], and that of the Rosenberg Self-Esteem Scale in Thai samples is 0.84, indicating good internal consistency [27]. The GPAs for the latest term of all students were obtained from the teachers' records. In all schools in Thailand, the maximum GPA is 4, and a GPA below 2.0 is considered a failing grade.

Data management and analysis

Data were collected with EpiData 3.1 software [28], and R software version 4.4.1 [29] was used for data analysis. Descriptive statistics were used to describe background characteristics and to estimate the magnitude of mental health problems, happiness, and self-esteem by using frequencies and percentages. Screen time in hours was reported as the mean and standard deviation (SD). Student's t-test was used to compare the difference between the two groups (screen time on weekdays and weekends) measured in a continuous scale. Chi-square tests (or Fisher's exact tests, where appropriate) and logistic regression were used to examine the factors associated with internalizing

problems as reported by teachers and parents. Any concerns reported by parents or teachers were reviewed and summarized by a psychologist into specific keywords. Subsequently, they were coded into four groups, including emotional, academic, internalizing, and externalizing problems, to use in further analysis. The GPAs of students in different grades were standardized. An academic problem was defined as a GPA < 2.5. Statistical significance was set at a P value < 0.05.

Ethical consideration

Written informed consent was obtained from the children's parents or caretakers. To prevent coercion, researchers conducted all data collection procedures themselves rather than asking teachers for help. To maintain autonomy, students could choose to respond to the questionnaire voluntarily, even if their parents had already provided consent. All participants could refuse or withdraw from the study at any time. They could complete or skip any questions. All data were kept confidential on a password-protected computer. Data were presented in aggregate form, not for an individual student. Ethical approval was obtained from the Ethics Review Committee at Prince of Songkla University (REC 62–091-18-1).

Results

Background characteristics of primary school children

A total of 2,499 students from nine schools were invited, and 2,277 students (91.1% response rate) participated in the study. The students were distributed proportionally to the sizes of the school types: five public schools (26.2%), two municipal schools (47.2%), and two private schools (26.6%). The median age [range]: 8 [6 – 12]. The male (n = 975) and female (n = 983) ratios were 1:1, and the distributions of the first, second, and third grades were 35.0%, 32.9%, and 32.1%, respectively.

Background characteristics of primary caretakers of primary school children

A total of 1,977 parents or caretakers completed the questionnaire (79.1% response rate). As shown in Table 1 , most respondents were Buddhist parents aged 31–40 years. Most of them had completed high school or higher. Most of them had blue-collar jobs or were housewives. Almost half of them reported incomes below 10,000 THB (1 USD = 32 THB).

Table 1 Sociodemographic data of parents and families of primary school children (N = 1,977) Sociodemographic data n (%) Gender Male 467 (24.0) Female 1480 (76.0) Relationship with the student Father/mother 1621 (83.3) Other relatives 326 (16.7) Age (years) ≤ 30 388 (20.1) >30 1544 (79.9) Religion Buddhist 1621 (83.9) Non-Buddhist 312 (16.1) Highest education level Below high school 478 (24.3) High school and above 1447 (74.7) Occupation Government officers/merchants/business 704 (36.5) Employee/housewife/others 1181(63.5) Family monthly income (THB) ≤ 10,000 777 (40.7) > 10,001 1134 (59.3) THB – Thai Baht (1 USD = 32 THB)

Family function

Parents reported that 34.2% had inadequate family functioning, while more than half reported that their family did not help each other solve problems, had neither adequate affective involvement nor an affective response, and had poor relationships and low trust in the family (Supplementary Table 1) .

Overall, respondents reported generally positive family functioning across most domains. The majority of items were rated as “moderately accurate” to “very accurate,” reflecting strong family cohesion, trust, affection, and effective problem-solving. Families commonly endorsed open communication, shared responsibilities, and mutual emotional support. However, several areas emerged as potential concerns. Some respondents reported difficulty expressing thoughts and feelings, uncertainty about family members’ reliability, and limited emotional expression within the household. A subset also indicated that family rules were unclear or too flexible, and that negative emotions were present within the family. These findings suggest that while most families demonstrate healthy functioning, specific aspects—particularly emotional openness, communication clarity, and consistency in family structure—may benefit from additional support or attention.

Game addiction and use of electronic devices

The assessment of game addiction reported by parents revealed 18.2% had an increased risk of game addiction, and 7.2% were considered to have game addiction already. One-fifth of the students had access to a mobile phone or had a desktop computer at home. Approximately half of them could connect to the internet at any time. The average screen time on weekdays was 3.5 h, rising to 6.2 h on weekends. Most time was spent watching television and using a mobile phone, followed by using a tablet or computer. They primarily used this information technology (IT) for entertainment (72.6%) and gaming (70.4%). They moderately used IT for learning/education (65.2%), schoolwork (57.5%), listening to music and watching movies (47.5%). Using IT for social connection (17.9%) and using email (7.8%) were less common for this age group (Supplementary Table 2) .

Table 2 Screen time on weekdays and weekends by use of information technology devices (ITs) (N = 1,977) Use of ITs Screen time (hrs) on weekdays Mean (SD) P value Screen time (hrs) on weekends Mean (SD) P value Own a mobile phone Yes 4.25 (3.26) < 0.001 7.90 (4.84) < 0.001 No 3.45 (3.02) 5.86 (3.80) Had desktop computers Yes 4.18 (3.51) < 0.001 7.02 (4.23) < 0.001 No 3.49 (2.97) 6.06 (4.04) Internet access Yes 3.97 (3.24) < 0.001 6.90 (4.2) < 0.001 No 3.18 (2.94) 5.47 (3.90) Types of devices used < 0.001 < 0.001 TV 2.25 (1.78) 3.54 (2.68) Mobile phone 2.04 (1.90) 3.06 (2.59) Computer 1.69 (1.47) 2.21 (1.90) Tablet 1.62 (1.08) 2.50 (2.14) Number of devices used < 0.001 < 0.001 1 2.50 (2.06) 3.98 (2.93) 2 4.17 (2.99) 6.48 (4.07) 3 5.87 (3.63) 7.58 (4.16) 4 6.59 (3.19) 8.78 (4.51) ITs – Information and technology; SD – Standard deviation; hrs - hours

Table 2 shows that average screen time on weekends was 1.5–2 times higher than on weekdays. School students who owned a mobile phone, had a desktop/laptop computer at home, and had unrestricted internet access significantly increased their screen time on both weekdays and weekends. Students spent most of their screen time watching television and using their mobile phones. The number of devices used significantly increased screen time in a dose■response manner.

Mental health problems

Table 3 Mental health problems based on DSM-5 screening questions responded by parents (N = 1,977) Mental health problems n (%) Any signs and symptoms suggested mental health problems before admission to kindergarten (n = 1,875) Yes (≥ 1 signs and symptoms could be chosen) 344 (18.3) • Suspected of attention deficit hyperactivity disorder 220 (63.9) • Social skills problems 109 (31.7) • Learning problems 130 (37.7) Any signs and symptoms suggested mental health problems after admission to kindergarten (n = 1,870) Yes (≥ 1 signs and symptoms could be chosen) 784 (41.9) • Learning problems 260 (31.1) • Social skills problems 185 (23.6) • Enuresis 148 (18.8) •

Soiling 46 (5.8) • Sleep problems 114 (14.5) • Hypersomnia 79 (10.0) • Parasomnia 346 (44.1) • Aggression and violence 185 (23.5) • Lies/stealing 90 (11.4)

According to the DSM-5 screening questions, 18.3% of parents reported having at least one mental health problem related to neurodevelopmental problems before admission to kindergarten. After they entered kindergarten, the overall number of issues more than doubled. Although learning and social skill problems slightly decreased, issues related to excretion, sleep, and negative behaviors were newly observed (Table 3).

In addition to DSM-5 screening, approximately 40% of either caretakers or teachers reported that a student had at least one problem. Figure 1 A and B show word clouds illustrating emotional, behavioral, and academic concerns reported by caretakers and teachers, respectively. Both groups most frequently identified learning difficulties, challenges in reading and writing, and attention deficit as prominent issues among children. Parents also highlighted oversensitivity, shyness, frustration, game overuse, and impulsivity, reflecting concerns about emotional regulation and social behavior. Teachers similarly emphasized academic problems but also noted absenteeism, hyperactivity, and medication use, suggesting greater awareness of classroom-related functioning and behavioral challenges.

Fig. 1 Perceived academic, emotional, and behavioral problems reported by parents (n = 579) and teachers (n = 211) A: Parents B: Teachers .

When all reported problems were categorized into four domains, academic issues were reported most frequently (76–78%) by both teachers and caregivers. Teachers reported that students had only academic problems, whereas caretakers additionally reported that at least 50% of students had difficulties in other emotional and behavioral domains. In contrast, the caretakers reported problems across multiple domains, including academic, emotional, and behavioral issues (Supplementary Table 3). Moreover, we found that students who had a moderate risk of game addiction (OR 5.61, 95% CI 1.91–17.42) and severe risk of game addiction (OR 7.98, 95% CI 1.96–29.4) were significantly associated with the internalizing behaviors observed by teachers (Supplementary Table 4) . In contrast, none of the students with high self-esteem reported problems similar to those mentioned by the teachers. Higher weekday screen time was significantly associated with increased internalizing behaviors reported by parents. Table 4 reported that almost all the students had moderate to high self-esteem. However, 10% felt sad to very sad toward themselves, and approximately one-third had poor to very poor life satisfaction. One-fourth of the students had GPAs less than 2.50.

Self-esteem, happiness, life-satisfaction, and academic achievement

Table 4 Self-esteem, happiness, life satisfaction completed by students and grade point average obtained from teachers (n = 2277) Parameters n (%) Level of self-esteem Low 8 (0.4) Moderate 1781 (77.8) High 483 (21.9) Happiness Feeling toward oneself Sad to very sad 237 (10.4) Neutral 363 (16.0) Happy to very happy 1668 (73.5) Life satisfaction Extremely dissatisfied to dissatisfied 719 (31.8) Neutral 566 (18.0) Satisfied to extremely satisfied 979 (43.1) GPAs (Grade point average) 3.00–4.00 1172 (49.7) 2.50-<3.00 575 (24.4) 2.00-<2.50 261 (11.1) <2.00 348 (14.8)

Discussion

This study provides important insights into the mental health problems and well-being of the first- to third-grade primary school students in southern Thailand. Approximately one-fourth of the students

reported low academic achievement, and one-third reported low life satisfaction. This pattern resonates with international findings that mental health difficulties are increasing among primary school children. A longitudinal study in Wales reported that the prevalence of emotional problems rose from 13 to 15% in 2017–2018 to 29% in 2021–2022 [30].

The relationship between academic achievement and mental health is bidirectional. In our study, one-fourth of the students had GPAs below 2.50, indicating that academic struggles may contribute to mental health difficulties. International studies confirm that poor mental health is associated with lower academic achievement [31 , 32]. Conversely, good mental health predicts better academic performance, underscoring the need to integrate academic and mental health support.

Additionally, the study identified high screen time as a critical factor associated with these rising mental health problems. Our students spent an average of 3.5 h of screen time per weekday and 6.2 h per weekend, with significantly higher use among those who owned mobile phones, had unrestricted internet access, or multiple devices at home. One-fourth of them were at risk, and 7.2% had already developed game addiction. These findings reflect global concerns about the impact of digital device use on children’s mental health. Excessive screen time has been consistently associated with anxiety, depression, behavioral issues, and sleep disturbances [33 – 35]. Excessive gaming has been linked to poor academic performance, emotional difficulties, and social isolation [36 – 38]. Parental supervision emerged as a key protective factor. Prior studies have shown that children with monitored screen use report fewer behavioral and emotional difficulties [39]. While digital technology can enhance learning and social connections, these benefits are realized only when use is balanced and guided.

Despite the presence of mental health problems, most students reported moderate to high self-esteem and levels of happiness. This balance of risk and resilience highlights the importance of strengthening protective factors while mitigating risks. Early identification of risk factors is essential, as these factors have been linked to a fourfold increased risk of later diagnosis of neurodevelopmental and psychiatric disorders [14]. Therefore, early childhood is a critical window for intervention, as unaddressed problems can escalate into more severe disorders during adolescence and adulthood.

Supportive family environments promote socioemotional development, academic achievement, and overall well-being. Peer relationships also play a critical role, with stronger connections associated with higher levels of happiness and academic performance, particularly among girls [40]. Identifying promotive and protective factors across multiple levels—individual, family, school, and community—is vital for addressing social, emotional, and behavioral difficulties [41].

Moreover, school well-being in the early years has been shown to predict continued well-being and peer relations later in life, suggesting that school-based prevention programs must be implemented early. Digital interventions also present opportunities for promoting mental health. School-based digital health promotion programs have been shown to improve knowledge, behaviors, and mental health outcomes, although evidence remains limited [42]. This highlights the potential of integrating digital approaches with traditional school-based interventions. Interventions rooted in positive psychology have shown promise. The Digital Netizen Alliance (D.N.A.) program, for example, has successfully reduced symptoms of gaming disorder and promoted healthier coping strategies [43]. These findings suggest that school- and community-based interventions can help manage problematic gaming while promoting digital literacy and balanced use.

The findings highlight several areas for targeted intervention. First, early identification of mental health problems and developmental delays is essential to prevent escalation into long-term disorders. Second, interventions addressing screen time and gaming behaviors must involve parents, emphasizing supervision and balanced use of technology. Third, schools should adopt whole-school approaches to mental health education, strengthen teacher capacity and promote home-school collaboration [44]. Structured programs such as the Treasure File Programme, while effective in enhancing self-esteem, must be adapted to address broader aspects of well-being [45]. Holistic approaches that incorporate individual, family, school, and community levels are most effective. This includes promoting positive relationships, building emotional literacy, and creating supportive school environments [46 , 47]. Importantly, not all interventions yield universal benefits, underscoring the need for tailored approaches that consider cultural and contextual factors. The large sample size of this study can yield precise estimates of the study outcomes. However, the cross-sectional design limits the ability to make causal inferences.

Conclusion

This study reveals a complex picture of primary school students' mental health in southern Thailand, where significant mental health challenges coexist with positive indicators of well-being, such as self-esteem and happiness. These findings are consistent with international evidence and emphasize the urgent need for comprehensive, multilevel interventions. Addressing risk factors such as screen time, gaming addiction, and poor peer relationships while reinforcing protective factors such as family support and school-based mental health programs is essential. A holistic, culturally sensitive approach involving parents, teachers, schools, and communities will be vital to promoting children's long-term mental health and well-being.

Supplementary Information

Below is the link to the electronic supplementary material.

Supplementary Material 1